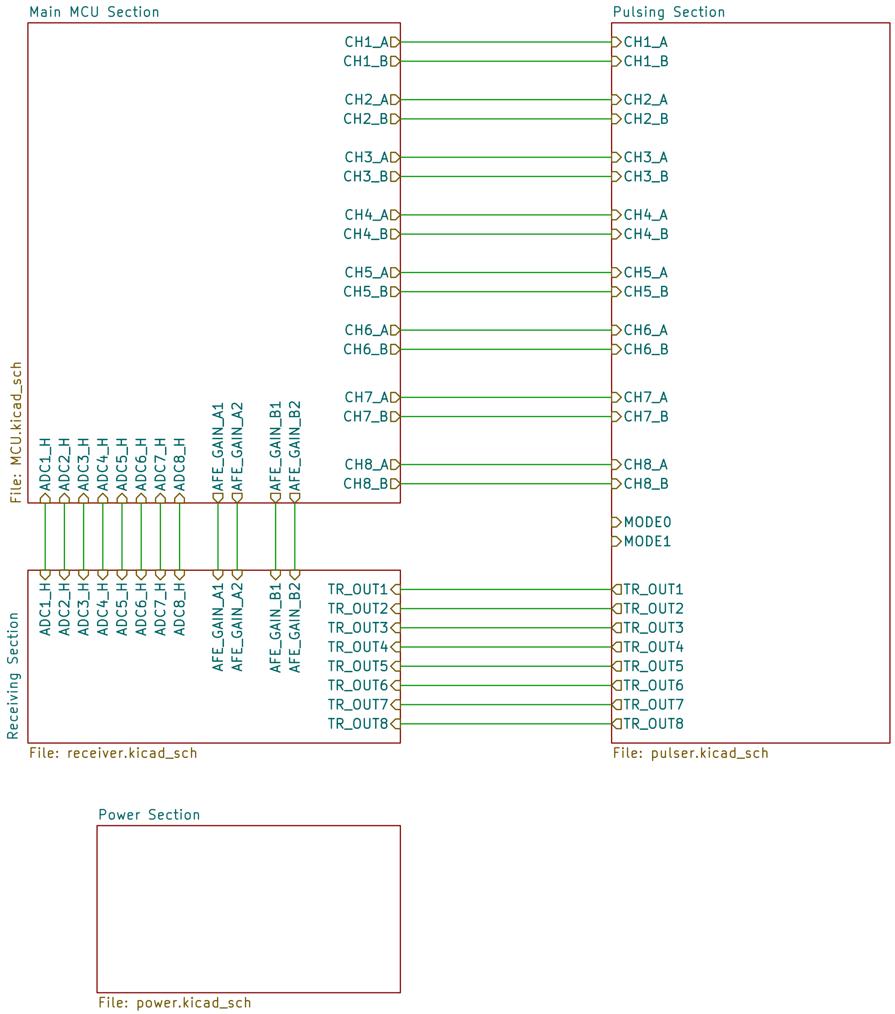

Main MCU Section
File: MCU.kicad_sch
CH1_A
CH1_B
CH2_A
CH2_B
CH3_A
CH3_B
CH4_A
CH4_B
CH5_A
CH5_B
CH6_A
CH6_B
CH7_A
CH7_B
CH8_A
CH8_B
ADC1_H
ADC2_H
ADC3_H
ADC4_H
ADC5_H
ADC6_H
ADC7_H
ADC8_H
AFE_GAIN_A1
AFE_GAIN_A2
AFE_GAIN_B1
AFE_GAIN_B2
Pulsing Section
MODE0
MODE1
TR_OUT1
TR_OUT2
TR_OUT3
TR_OUT4
TR_OUT5
TR_OUT6
TR_OUT7
TR_OUT8
File: pulser.kicad_sch
Receiving Section
File: receiver.kicad_sch
Power Section
File: power.kicad_sch

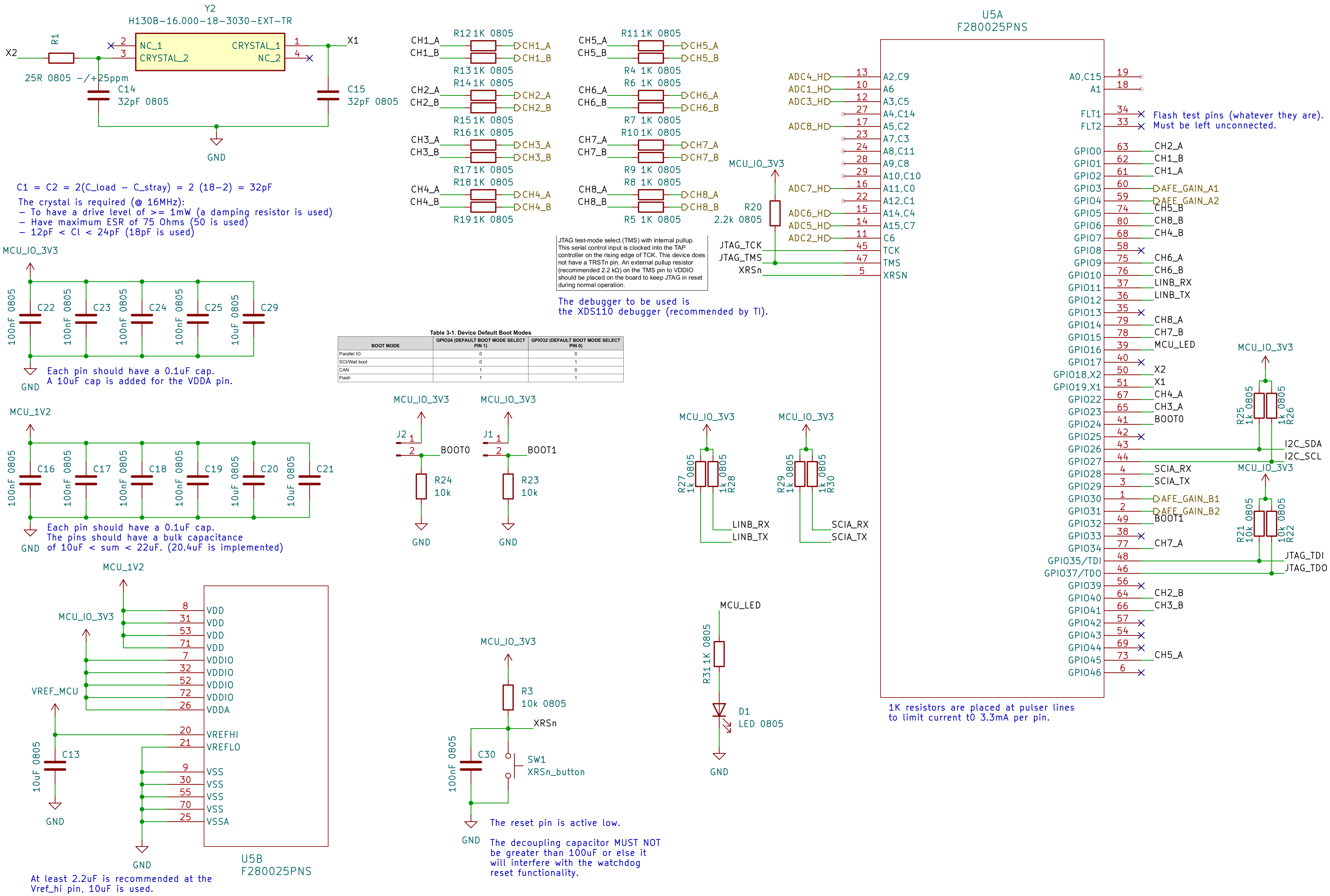
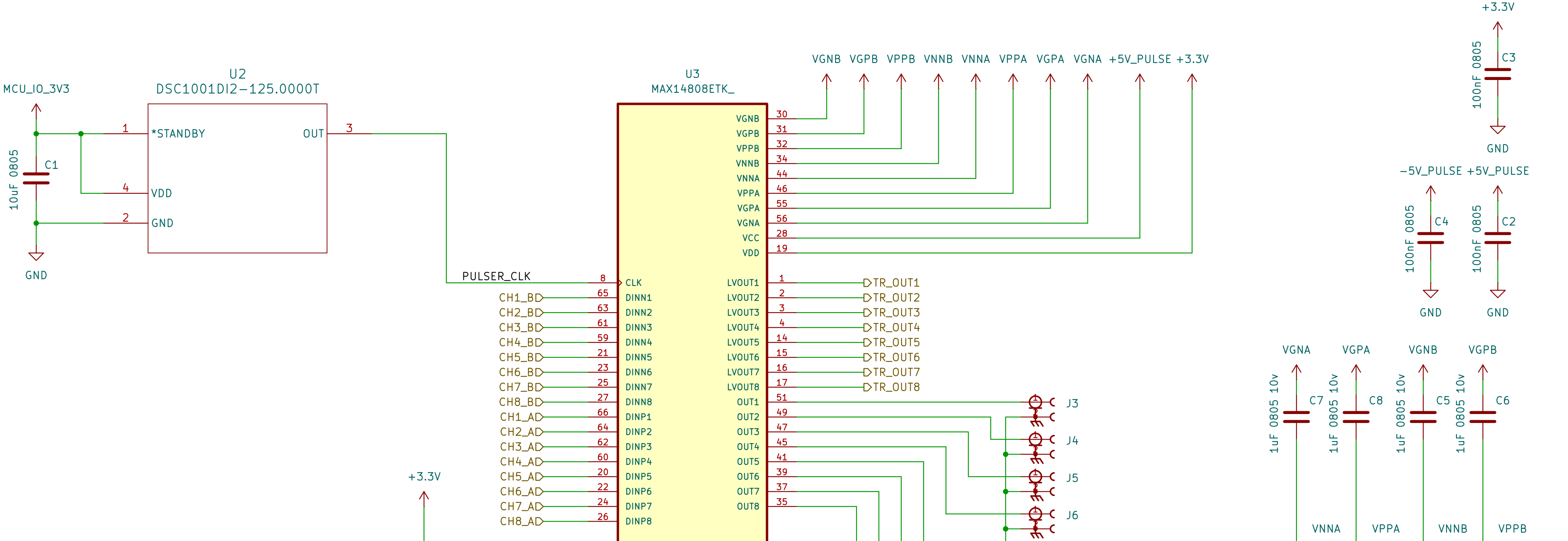
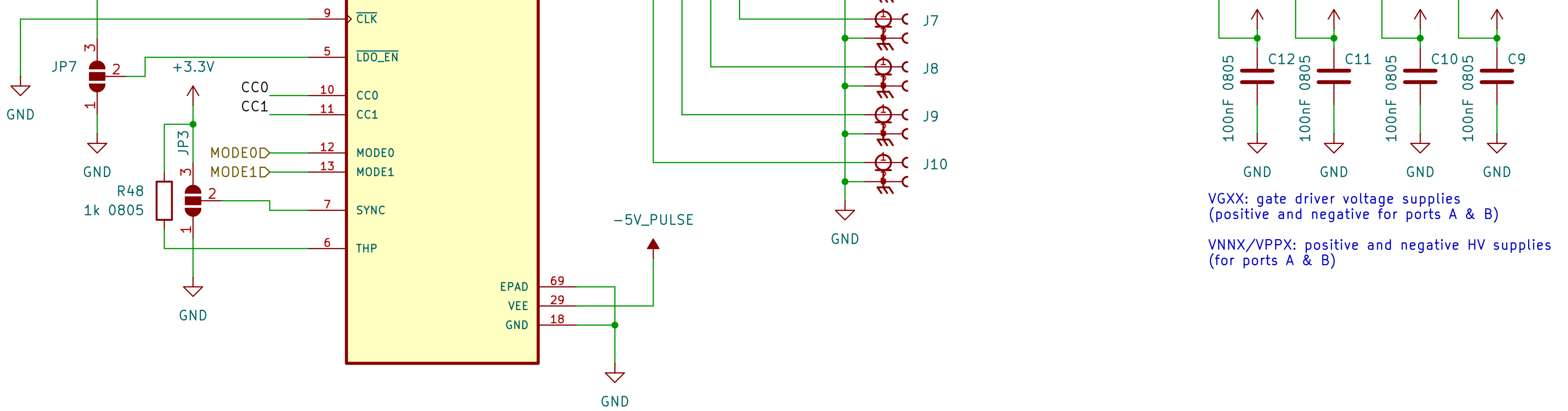


Table 3-1. Device Default Boot Modes

BOOT MODE	GPIO24 (DEFAULT BOOT MODE SELECT PIN 1)	GPIO32 (DEFAULT BOOT MODE SELECT PIN 0)
Parallel IO	0	0
SCI/Wait boot	0	1
CAN	1	0
Flash	1	1



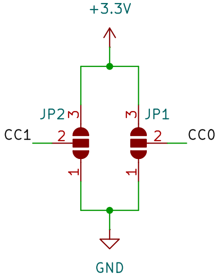


VGXX: gate driver voltage supplies (positive and negative for ports A & B)

VNNX/VPPX: positive and negative HV supplies (for ports A & B)

Table 5. Current Drive Selection

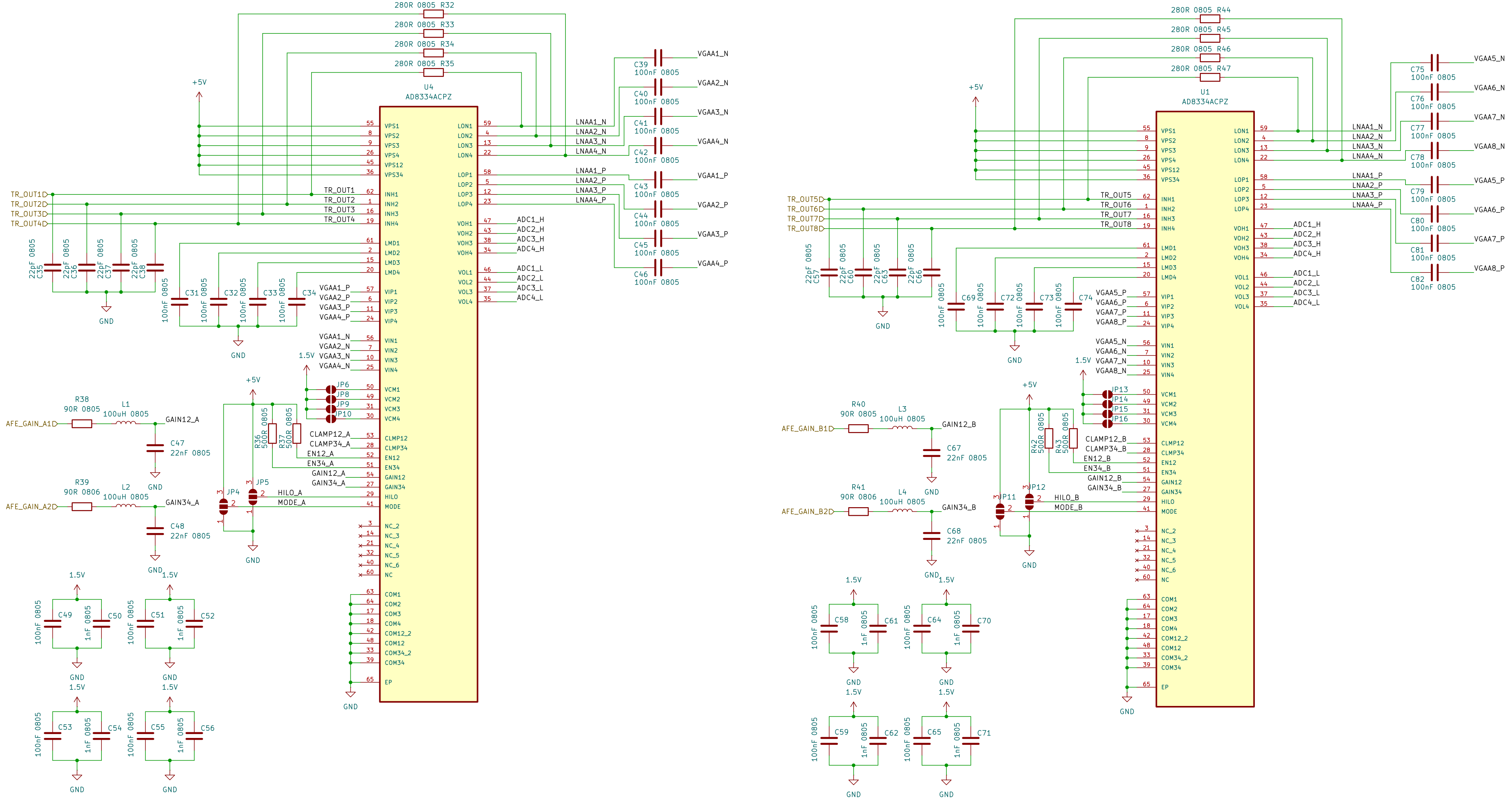
INPUTS		PULSER OUTPUT CURRENT (typ)
CC0	CC1	
0	0	2A
1	0	1.5A
0	1	1A
1	1	0.5A



Enabling/Disabling the internal floating power supplies (FPS):

requirements in most of the cases. Drive LDO_EN low or leave unconnected to enable the internal FPSs; drive LDO_EN high to disable the internal FPSs.

A triple-option solder jumper is placed to either pull-up (select external FPS) or pull-down (select internal FPS) the LDOEN pin.



LNA Current					
AD8331 (ENBL)	Each channel	7.5	11	15	mA
AD8332, AD8334 (ENBL)	Each channel	7.5	12	15	mA
VGA Current					
AD8331 (ENBV)		7.5	14	20	mA
AD8332, AD8334 (ENBV)		7.5	17	20	mA

0.5k would draw 10mA, acceptable.

Table 7. LNA External Component Values for Common Source Impedances

R_{IN} (Ω)	R_{IZ} (Nearest STD 1% Value, Ω)	C_{SH} (pF)
50	280	22
75	412	12
100	562	8
200	1.13 k	1.2
500	3.01 k	None
6 k	∞	None

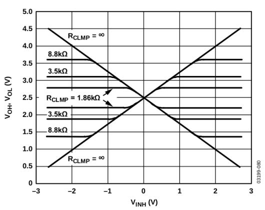
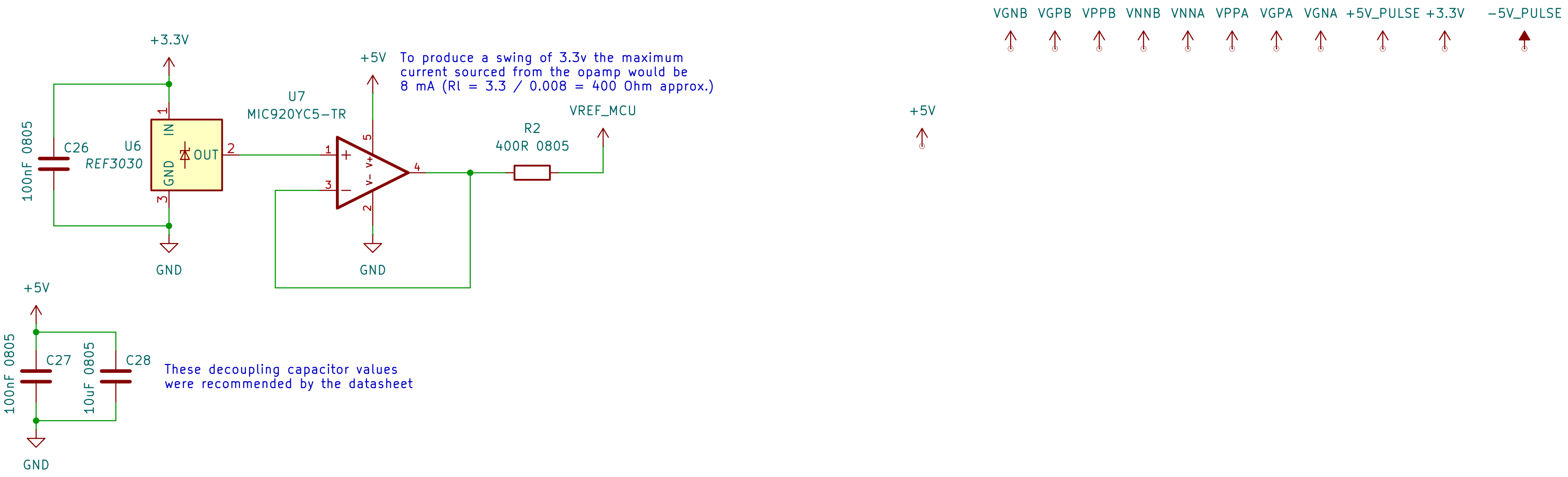


Figure 80. Output Clamping Characteristics

2nd order filter values were recommended
by a TI application note (spraa88a), specs:
- PWM frequency = 5 MHz
- Cut-off frequency = 107.3 KHz
- Damping ratio = 0.675 (over-damped but response time = 3 uSec)

MODE INTERFACE (PIN MODE)			
Logic Level for Positive Gain Slope	0	1.0	V
Logic Level for Negative Gain Slope	2.25	5	V
HILO GAIN RANGE INTERFACE (PIN HILO)			
Logic Level to Select HI Gain Range	2.25	5	V
Logic Level to Select LO Gain Range	0	1.0	V


+3.3V
100nF 0805
C26
U6
REF3030
IN
OUT
GND
GND
+5V
U7
MIC920YC5-TR
To produce a swing of 3.3v the maximum
current sourced from the opamp would be
8 mA (Rl = 3.3 / 0.008 = 400 Ohm approx.)
GND
VREF_MCU
R2
400R 0805
+5V
100nF 0805
C27
10uF 0805
C28
These decoupling capacitor values
were recommended by the datasheet
GND
+5V
VGNB
VGPB
VPPB
VNNB
VNNA
VPPA
VGPA
VGNA
+5V_PULSE
+3.3V
-5V_PULSE